PRE-CLASS READING · FINANCIAL STATEMENT ANALYSIS

DuPont Analysis and the Altman Z-score

Concepts, methods and a guide to the six cases

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Why read this before class? A board that sees a return on equity of 18 per cent asks two questions. Where did the 18 per cent come from? And would the firm still earn it if conditions turned? DuPont analysis answers the first by splitting the return into parts a manager can act on. The Altman Z-score answers the second by combining five ratios into one early-warning number. This note explains both tools, then shows the six techniques you need for the case set. Every technique is worked once on an invented firm, so when you open a case you will know which formula to use, in what order, and what to watch.

Case	What you must do	Where to learn it
D1 Kaveri Appliances	Split the change in ROE with the 3-way and 5-way forms; measure each driver's contribution	Sections 1.2 and 1.3 (Technique A)
D2 Aurum and Bazaar	Find the ROE engine; test a margin squeeze	Section 1.4 (Technique B)
D3 Sunrise Textiles	Separate headline ROE from core ROE	Section 1.5 (Technique C)
Z1 Deccan Auto	Read the zones; break the change in Z into weighted terms	Sections 2.1 to 2.3 (Technique D)
Z2 Marine Chemicals	Price dependence, break-even price, Z″ check	Section 2.4 (Technique E)
Z3 Gokul Engineering	Choose the right model; Z′ and balance-sheet fixes	Section 2.5 (Technique F)

Colour guide. Blue boxes hold key concepts; green boxes hold formulas and steps; orange boxes show applications; red boxes warn about challenges and traps; purple boxes give insights; teal boxes give the manager's lens. In the diagrams, blue is profit margin, orange is asset efficiency, purple is leverage and teal is tax and interest.

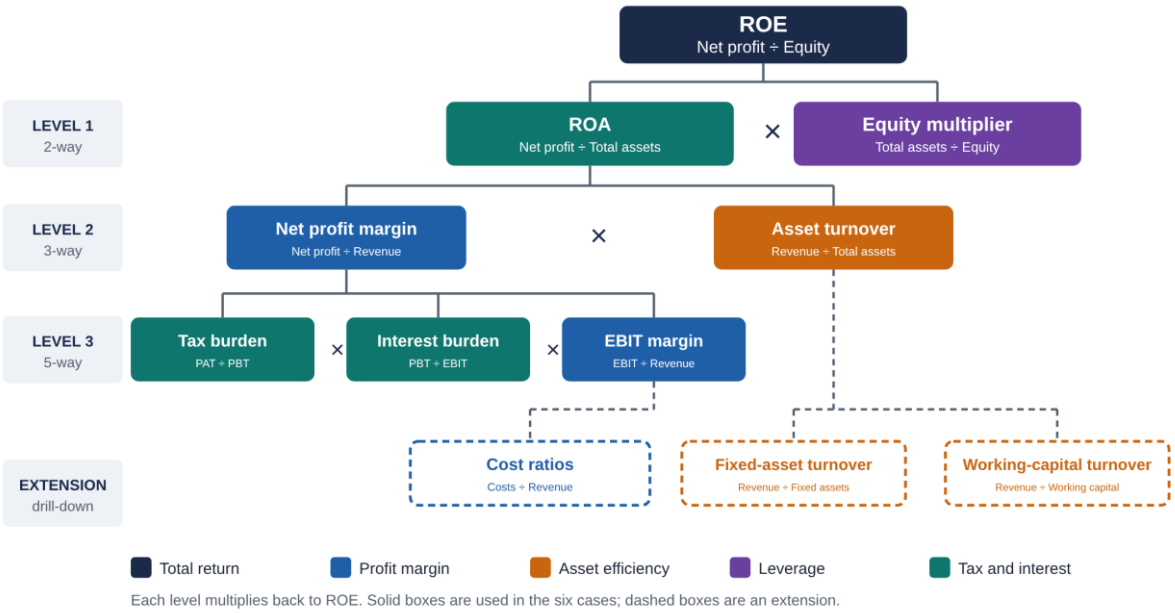
KEY CONCEPT

One running example. Sundaram Auto Components Ltd is an invented listed manufacturer. In Year 1 its return on equity is 12.00 per cent. In Year 2 it borrows ₹150 crore to buy back shares and ROE rises to 13.84 per cent. Section 1 explains why, and Section 2 shows how its Z-score changes. Amounts are in ₹ crore. The case companies are different firms, so nothing here gives away a case answer.

1. DuPont analysis

1.1 What it is, why and when

DuPont analysis breaks the return on equity (ROE) into the decisions that create it. Profit per rupee of equity depends on how much profit a rupee of sales earns, how many rupees of sales each rupee of assets produces, and how much of the assets the owners finance. Use it in a performance review, a peer comparison, a target-setting exercise, a credit or investment screen, or whenever ROE moves and the board asks why.



1.2 The versions: from two links to five

The tree above is one identity read at different depths. Each level multiplies back to ROE, so you can stop at the depth your question needs.

Level	Formula	Answers the question	Cases
2-way	ROE = ROA × Equity multiplier	Is the return from operations or from borrowing?	D1 (check)
3-way	ROE = Net profit margin × Asset turnover × Equity multiplier	Margin, efficiency or leverage?	D1
5-way	ROE = Tax burden × Interest burden × EBIT margin × Asset turnover × Equity multiplier	Which part of the margin moved: operations, interest or tax?	D1, D2, D3
Extension	Asset turnover into fixed-asset and working-capital turnover; EBIT margin into cost ratios	Which asset or cost line is behind the change?	Not needed in the six cases

FORMULA AND RULE

Definitions used in the cases (year-end balances): Tax burden = PAT ÷ PBT. Interest burden = PBT ÷ EBIT. EBIT margin = EBIT ÷ Revenue. Asset turnover = Revenue ÷ Total assets. Equity multiplier = Total assets ÷ Equity. ROA = PAT ÷ Total assets.

How the levels link: Net profit margin = Tax burden × Interest burden × EBIT margin. ROA = Net profit margin × Asset turnover. ROE = ROA × Equity multiplier.

Always check: the product of the ratios must equal PAT ÷ Equity. If it does not, a ratio is wrong.

1.3 Technique A: explain a change in ROE (Case D1)

When ROE changes, the question is how much of the change each driver caused. Because the drivers multiply, their changes cannot simply be subtracted. The log method splits the change so that the contributions add up exactly.

FORMULA AND RULE

Steps

1. Compute the ratios for both years and confirm that the product equals PAT ÷ Equity.
2. Find $\Delta ROE = ROE_2 - ROE_1$, in percentage points.
3. For each driver: Contribution = $\Delta ROE \times \ln(\text{driver}_2 \div \text{driver}_1) \div \ln(ROE_2 \div ROE_1)$.
4. Check that the contributions add up to ΔROE. In the 5-way form the three margin drivers add up to the net-profit-margin contribution of the 3-way form.
5. Read the result: ROA drivers show the operating story; the equity multiplier shows how much leverage helped.

Worked on Sundaram. ROE moved from 12.00% to 13.84%, so ΔROE = +1.84 points and $\ln(ROE_2 \div ROE_1) = 0.1427$.

Driver	Year 1	Year 2	3-way (pp)	5-way (pp)
Net profit margin	6.00%	5.19%	−1.87	
Tax burden	0.720	0.720		+0.00
Interest burden	0.833	0.721		−1.87
EBIT margin	10.00%	10.00%		+0.00
Asset turnover	1.000×	1.000×	+0.00	+0.00
Equity multiplier	2.000×	2.667×	+3.71	+3.71
Total = ΔROE			**+1.84**	**+1.84**

INSIGHT

Operations did not improve: EBIT margin and asset turnover are unchanged, and the extra interest cost pulled ROE down by 1.87 points. The whole rise, and more, came from the equity multiplier (+3.71 points). A leverage-led rise in ROE like this is a warning that returns improved because owners put in less equity, not because the business earned more. Case D1 asks you to make exactly this judgement when growth is funded by debt.

CHALLENGE AND ISSUE

- Use the same balance basis in both years. The cases use year-end balances; say so in your answer.
- Keep the sign. A negative contribution takes ROE down.
- Do not compute contributions from rounded ratios; carry full figures and round at the end.
- Margin contribution in the 3-way table equals the sum of the three margin lines in the 5-way table. A small gap is rounding.

1.4 Same ROE, different engine (Technique B, Case D2)

Two firms can show the same ROE and have very different risk. The engine test asks which ratio is most unusual compared with the other firm. A very high margin means margin-led, a very high turnover means turnover-led, and a very high equity multiplier means leverage-led.

The squeeze test then asks what happens to ROE when EBIT margin falls by one percentage point, with revenue, interest, tax rate, assets and equity unchanged.

Firm (invented)	EBIT margin	Asset turnover	Equity multiplier	ROE	Engine
Annapurna Retail Ltd	5.3%	3.00×	2.00×	18.0%	Turnover-led
Veda Labs Ltd	21.3%	0.80×	1.50×	18.0%	Margin-led
Rudra Steel Ltd	12.0%	1.00×	3.00×	18.0%	Leverage and turnover

FORMULA AND RULE

Squeeze steps. $EBIT_{ne^{\chi}} = EBIT - 1\% \times Revenue$. $PBT_{ne^{\chi}} = EBIT_{ne^{\chi}} - Interest$. $PAT_{ne^{\chi}} = PBT_{ne^{\chi}} \times (1 - \text{tax rate})$. $ROE_{ne^{\chi}} = PAT_{ne^{\chi}} \div Equity$. Then compare the fall in ROE and the new interest cover (EBIT ÷ Interest).

Firm	ROE before	ROE after squeeze	Fall (pp)	Fall (% of ROE)	Interest cover before / after
Annapurna Retail Ltd	18.0%	13.5%	4.50	25.0%	4.0× / 3.3×
Veda Labs Ltd	18.0%	17.1%	0.90	5.0%	17.0× / 16.2×
Rudra Steel Ltd	18.0%	15.8%	2.25	12.5%	3.0× / 2.8×

INSIGHT

All three firms earn 18.0 per cent, yet the same one-point cost shock cuts their ROE by very different amounts. The thinner the margin on a given revenue, and the bigger the interest bill, the more of the profit one point of cost removes. **Equal ROE does not mean equal risk.** The DuPont profile tells an investor which risk comes with the return.

1.5 Headline ROE versus core ROE (Technique C, Case D3)

A one-off gain, such as a land sale, lifts profit for a year. To judge performance, remove the gain and the tax on it, then recompute ROE and the margins. Do not remove the gain and forget its tax.

FORMULA AND RULE

Steps. 1. Compute reported ROE and its 5-way ratios. 2. Core EBIT = Reported EBIT − Exceptional gain. 3. Core PBT = Core EBIT − Interest. 4. Core PAT = Core PBT × (1 − tax rate). 5. Core ROE = Core PAT ÷ Equity; core EBIT margin = Core EBIT ÷ Revenue. 6. Compare reported with core, and each with last year.

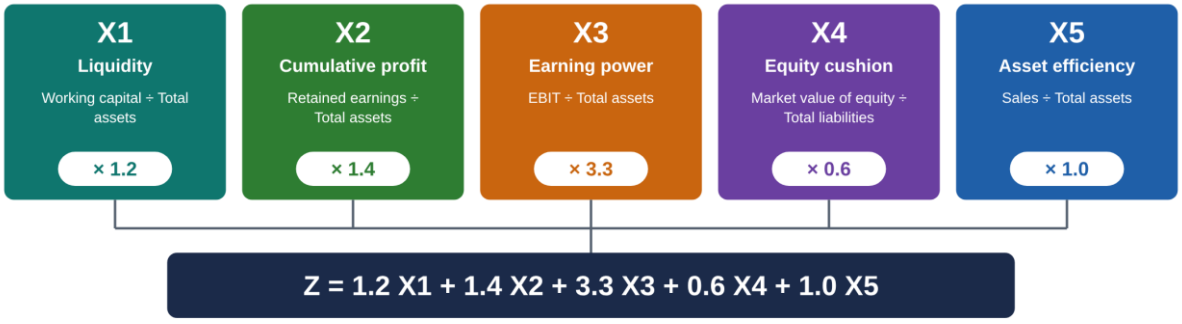
Tara Foods Ltd (invented)	Reported	Core	Last year ROE
Operating profit / Exceptional gain	100 / 24	100 / 0	
Interest	12	12	
PBT	112	88	
PAT at 25% tax	84	66	
Equity	500	500	
ROE	**16.8%**	**13.2%**	13.0%

> **INSIGHT**
> Reported ROE jumped 3.8 points, but core ROE moved only 0.2 points. The improvement was the land sale, not the business. An investor who screens on reported ROE alone buys a one-off. Case D3 asks you to prepare the same comparison for an investment committee.

2. The Altman Z-score

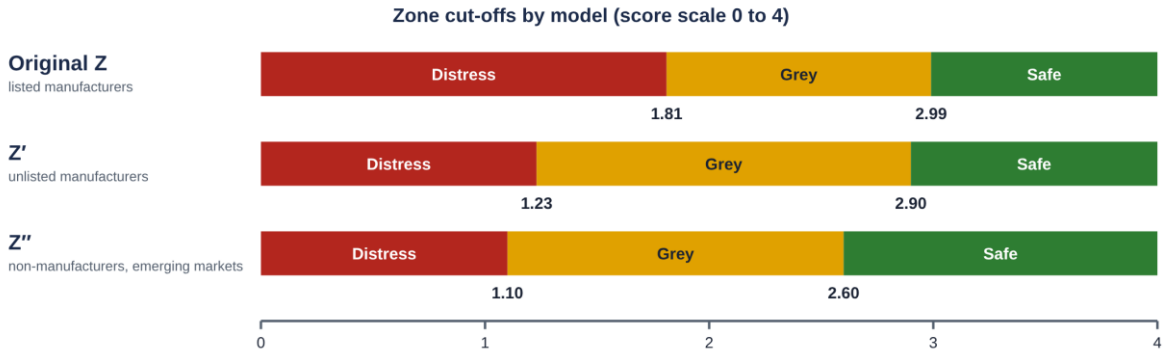
2.1 What it is, why and when

In 1968 Edward Altman combined five ratios into a single score using discriminant analysis. He fitted it on 66 manufacturers, half of which later failed. The weights show how strongly each ratio separated failed firms from healthy ones. A low score puts a firm with the failures, a high score with the survivors, and the middle is a grey area. Use it before extending credit or trade credit, when a covenant or rating is at risk, when screening an acquisition target, and when tracking a turnaround.



The weights show how strongly each ratio separated failed from healthy firms. X3 (earning power) carries the most weight.

2.2 Three models, three sets of zones



Model	Formula	Zones	Use for
Original Z	1.2 X1 + 1.4 X2 + 3.3 X3 + 0.6 X4 + 1.0 X5. X4 = market value of equity ÷ total liabilities	2.99 / 1.81	Listed manufacturers
Z′	0.717 X1 + 0.847 X2 + 3.107 X3 + 0.420 X4 + 0.998 X5. X4 = book equity ÷ total liabilities	2.90 / 1.23	Unlisted (private) manufacturers
Z″	6.56 X1 + 3.26 X2 + 6.72 X3 + 1.05 X4. X4 = book equity ÷ total liabilities. Emerging-market form adds a constant of 3.25	2.60 / 1.10	Non-manufacturers, services, emerging markets

CHALLENGE AND ISSUE

- Use the right weights **and** the right zones for the model you chose. Mixing Z weights with Z′ zones is a common error.
- X1 is (current assets − current liabilities) ÷ total assets in the cases. X4 in the original Z uses **market capitalisation**; in Z′ and Z″ it uses **book equity**.
- Z″ has no sales term (no X5).
- The score is a classification, not a probability of default. Banks, NBFCs and insurers do not fit these models.

2.3 Technique D: read the zones and break the change in Z into terms (Case Z1)

FORMULA AND RULE

Steps

1. Compute X1 to X5 for each year with the right definitions.
2. Multiply each ratio by its weight to get the five **weighted terms**.
3. Add the terms to get Z and read the zone from the cut-offs.
4. Change in Z = sum of the changes in the five terms. Show each term's change to see which one moved the score.
5. **EBIT needed for the safe zone:** $\Delta X3 = (2.99 - Z) \div 3.3$ and $\Delta EBIT = \Delta X3 \times$ Total assets (other ratios held constant).

Term (Sundaram)	Year 1	Year 2	Change
X1 Liquidity	0.150 × 1.2 = 0.180	0.120 × 1.2 = 0.144	−0.04
X2 Retained earnings	0.250 × 1.4 = 0.350	0.125 × 1.4 = 0.175	−0.17
X3 EBIT / assets	0.100 × 3.3 = 0.330	0.100 × 3.3 = 0.330	+0.00
X4 Market equity / liabilities	2.000 × 0.6 = 1.200	1.333 × 0.6 = 0.800	−0.40
X5 Sales / assets	1.000 × 1.0 = 1.000	1.000 × 1.0 = 1.000	+0.00
Z and zone	**3.06: Safe**	**2.45: Grey**	**−0.61**

INSIGHT

Z falls from 3.06 to 2.45, from safe to grey. The largest fall is in X4 (the market value of equity against liabilities), then X2 and X1, while the earning-power term (X3) does not change at all. For Sundaram to get back to 2.99 on EBIT alone, EBIT would have to rise by about ₹197 crore, against ₹120 crore today, so a better score depends on rebuilding the balance sheet and the share price as well. Case Z1 asks for the same working for a borrower whose limit is up for renewal.

2.4 Technique E: how much of Z is the share price? (Case Z2)

Only X4 depends on the share price. When the price falls faster than the business, Z drops even though the fundamentals have not changed. Two checks separate the market signal from the business signal.

FORMULA AND RULE

Steps. 1. Hold X1, X2, X3 and X5 fixed and recompute X4 at each price (Market cap = Price × Shares). 2. **Break-even price for the safe zone:** $0.6\,X4 = 2.99 - (1.2\,X1 + 1.4\,X2 + 3.3\,X3 + 1.0\,X5)$; Market cap = X4 × Total liabilities; Price = Market cap ÷ Shares. Repeat with 1.81 to see whether distress is reachable on price alone. 3. **Book-value check:** recompute with Z″ using book equity ÷ total liabilities and its zones (2.60 and 1.10).

INSIGHT

Sundaram has 20 crore shares, so its market capitalisation of ₹1000 crore is a price of ₹50. Here Z = 2.45. The safe zone needs a price of about ₹83.81. Distress would need the price to fall below about ₹10.06. On the book-value check, Z″ = 2.50 (grey, cut-offs 2.60 and 1.10). Price and book value tell a consistent story here. In Case Z2 they do not, which is the point of the exercise.

2.5 Technique F: choose the model; balance-sheet fixes and Z′ (Case Z3)

Is the firm…	Use	Note
Listed, manufacturing	Original Z	X4 needs the market price
Unlisted, manufacturing	Z′	No market price exists; X4 uses book equity
Service, non-manufacturing, emerging market	Z″	Drops X5; adds 3.25 for emerging markets
Bank, NBFC or insurer	None of these	Use sector measures

A lender often proposes balance-sheet fixes: repay debt, convert debt to equity, sell idle assets. To test a proposal, recompute total assets, total liabilities, equity and working capital after the change, then recompute every ratio, because one change can move several ratios. Compare the fix with an operating improvement, whose weight is much larger.

INSIGHT

Suppose Sundaram were unlisted. Its Z′ would be 1.75 (grey). Converting ₹150 crore of debt into equity lifts Z′ to 1.92, a gain of 0.17, because only X4 moves and carries a weight of 0.420. The same gain would need EBIT to rise by about ₹65 crore, because X3 carries a weight of 3.107. **Balance-sheet engineering moves Z′ less than earnings do.** Case Z3 tests three such proposals and asks for a recommendation.

3. Solving the six cases

Case	Do this, in order	Common trap	Decision asked
D1	3-way and 5-way ratios for both years; check the product; contributions by the log method; evaluate the claim	Using rounded ratios; losing the sign; mixing balance bases	Is leverage hiding weaker operations?
D2	5-way for both firms; name the engine; apply the 1-point EBIT squeeze; compare ROE fall and interest cover	Cutting net margin instead of EBIT margin; forgetting interest stays fixed	Which firm suits a risk-averse investor?
D3	Reported 5-way; contributions; remove the gain and its tax; core ROE and margin	Removing the gain but not its tax	Is the improvement real?
Z1	X1 to X5 both years; weighted terms; Z and zone; term-by-term change; EBIT needed for 2.99	Wrong X1 or X4 definition; using the wrong zone	Renew the limit, and on what conditions?
Z2	Z at each price; break-even price; Z″ with book equity	Using Z weights in the Z″ check; forgetting that only X4 changes	Downgrade the credit or only watch it?
Z3	Pick the model; base score; each proposal alone; all together	Using original Z; ignoring that repaying debt shrinks assets	Approve the loan, and with what safeguards?

MANAGER'S LENS

Before you write the answer

- Name the model or version you used and the balance basis (year-end).
- State the formula once, then show the working in a table with both years.
- Check the identity (DuPont product equals PAT ÷ Equity; Z terms add to Z).
- End with the decision, in two or three sentences, and say what you would monitor next.
- Read DuPont and Z together: a higher ROE with a lower Z-score means the return was bought with risk.

Self-check before the cases

SELF-CHECK (ANSWERS IN ITALICS)

Q1. A firm has EBIT margin 10%, interest burden 0.80, tax burden 0.70, asset turnover 1.5 and equity multiplier 2.0. What is ROE? *0.70 × 0.80 × 10% × 1.5 × 2.0 = 16.8%.*

Q2. In the log method, ΔROE is 2.0 points and $\ln(ROE_2 \div ROE_1)$ is 0.10. A driver has $\ln(driver_2 \div driver_1) = 0.05$. Its contribution? *2.0 × 0.05 ÷ 0.10 = +1.0 point.*

Q3. An unlisted manufacturer needs a distress screen. Which model and which zone for safe? *Z'; above 2.90.*

Q4. Z = 2.30 on the original model, total assets ₹1,000 crore. EBIT must rise by how much to reach 2.99, other ratios fixed? *(2.99 − 2.30) ÷ 3.3 × 1,000 = ₹209 crore.*

Sources checked. E. I. Altman, Journal of Finance (1968), as summarised by the CFA Institute (2016) and NYU Stern; model weights and zones checked against Wikipedia, Wall Street Prep and calculate.co.nz; DuPont framework from FastGraphs and standard texts. Check weights and zones against the source you cite.